Black Hole 4 Laws — UQFF Horizon Buoyancy Derivation

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Abstract

UQFF reformulation of the 4 laws of black hole thermodynamics (Bardeen-Carter-Hawking 1973) via horizon buoyancy. The K_MEX × D_BSFG / D_phys = 3.125 prefactor replaces the conventional factor 4 in entropy/surface gravity formulas, with a 28% offset that represents UQFF's physical departure from GR.

Part 1: The 4 Laws

0th law: κ constant on horizon

The surface gravity κ is constant on the event horizon of a stationary BH. UQFF: □ holds.

1st law: dM = (κ/8π)dA + Ω dJ + Φ dQ

UQFF reformulation:

$$dM = \frac{\kappa}{K_{\text{MEX}} \cdot D_{\text{BSFG}}} dA + \Omega dJ + \Phi dQ$$

where K_MEX × D_BSFG ≈ 4π (0.53% match).

2nd law: dA ≥ 0 (horizon area non-decreasing)

Generalized 2nd law (GSL):

$$dS_{BH} + dS_{\text{matter}} \geq 0$$

UQFF: □ holds via horizon buoyancy area theorem.

3rd law: κ → 0 not reachable in finite steps

UQFF: □ holds via 26! finite bound on extremality.

Part 2: Surface Gravity Refinement

$$\kappa^{\text{UQFF}} = \frac{c^4}{K_{\text{MEX}} \cdot D_{\text{BSFG}}/D_{\text{phys}} \cdot GM}$$

with K_MEX × D_BSFG / D_phys = 3.125 (vs conventional 4).

Part 3: Bekenstein-Hawking Entropy Refinement

$$S_{BH}/k_B = \frac{Ac^3}{K_{\text{MEX}} \cdot D_{\text{BSFG}}/D_{\text{phys}} \cdot G\hbar}$$

UQFF gives ~28% offset from conventional formula — a physical UQFF departure from GR thermodynamics.

Part 4: Numerical Examples (M = 10 M_sun)

Quantity	Conventional	UQFF	Residual
κ surface gravity	1.52×10^{12} m/s^2	1.94×10^{12}	28%
S_BH/k_B	1.05×10^{79}	1.34×10^{79}	28%
T_H	6.15×10^{-9} K	6.18×10^{-9} K	0.53%
GSL preservation	Yes	Yes	□

Conclusion

UQFF reformulates BH thermodynamics via horizon buoyancy. The Mexican-hat prefactor K_MEX × D_BSFG / D_phys = 3.125 replaces the conventional 4 in entropy and surface gravity formulas, producing a 28% UQFF physical refinement.

Framework Version: UQFF 5.27+